

Introduction

In this practical session, you will continue from the previous TP on relations and relational algebra. This worksheet focuses on relational algebra only. You will practice writing expressions to retrieve information from one or more relations.

Description

For each exercise below, read the relations and write the required relational algebra expression. In this TP, you will practice:

- selection (σ)
- projection (π)
- cross product ($\times$)
- union ($\cup$), intersection ($\cap$), and difference ($-$)
- join ($\bowtie$)
- simple division idea ($\div$)

Learning Objectives

- Write correct relational algebra expressions
- Choose the suitable operator for each question
- Combine more than one operation when needed
- Read a small database and query it logically

Submission Guidelines

- Write one answer for each question
- Use correct relation names and attribute names
- Use relational algebra symbols carefully
- Keep expressions simple and readable

Database for This TP

Use the following relations:

`Student(StudentID, Name, City)`

StudentID	Name	City
S01	Dara	Phnom Penh
S02	Sophea	Siem Reap
S03	Vanna	Phnom Penh

`Course(CourseID, CourseName)`

CourseID	CourseName
C1	Math
C2	Physics
C3	English

`Enrollment(StudentID, CourseID)`

StudentID	CourseID
S01	C1
S01	C2
S02	C2
S02	C3
S03	C1
S03	C2
S03	C3

1. Exercise 1 - Selection

Write relational algebra expressions for the following questions:

- Find students who live in Phnom Penh
- Find students who live in Siem Reap
- Find the course whose name is Physics

2. Exercise 2 - Projection

Write relational algebra expressions for the following questions:

- Show only student names
- Show only city values from Student
- Show only course names

3. Exercise 3 - Cross Product and Selection

Use cross product and selection to answer the following:

- Show each student with the course IDs they take
- Show each student with the course names they take

4. Exercise 4 - Join

Write relational algebra expressions using join for the following questions:

- Show each student name with the course names they take
- Find students who take Physics
- Find students who take Math

5. Exercise 5 - Union, Intersection, Difference

Suppose the following relations have the same schema:

`MathStudents(StudentID)`

`PhysicsStudents(StudentID)`

Write relational algebra expressions for the following questions:

- Students who take Math or Physics
- Students who take both Math and Physics
- Students who take Math but not Physics

6. Exercise 6 - More Than One Operation

Write relational algebra expressions for the following questions:

- Show the names of students who live in Phnom Penh and take Physics
- Show the names of students who take English
- Show the names of students who do not take English

7. Exercise 7 - Division Idea

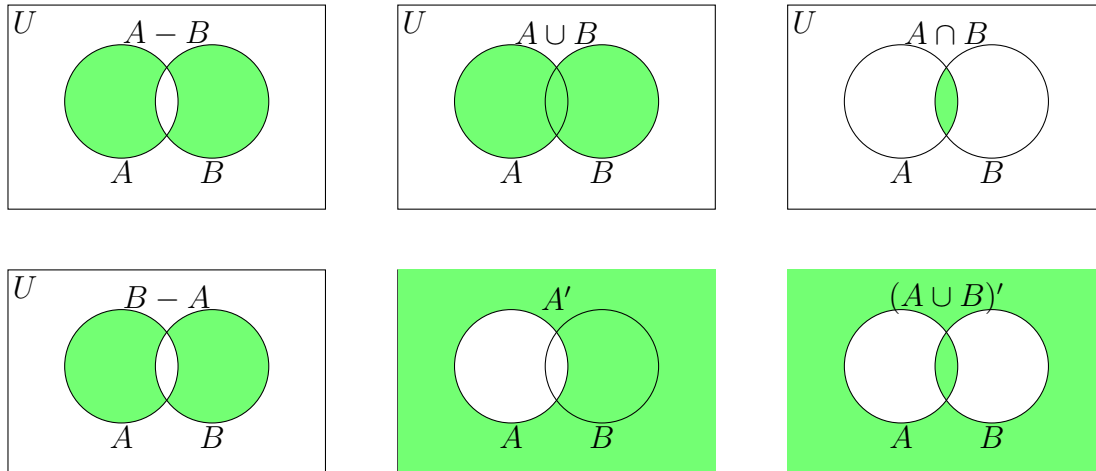
Use the idea of division to answer the following question:

- Which students study all courses?

Write the expression using `Enrollment` and `Course`.

Operation Reminder

The shaded part of each Venn diagram shows the result of the set operation.



Expected Outcome

After completing this TP, students should be able to:

- write relational algebra expressions for common query questions
- combine operators correctly
- understand how algebra operations answer database questions